
Watch, Listen, Read: A Controlled Multimodal Ablation for Hate Video Classification on HateMM

Youssef Dib Kevin Abou Jaoude Rami Aschkar

Group 7

Abstract

We study how much fusing three modalities (visual frames, audio, and ASR transcripts) improves hate video classification over each modality alone on the HateMM benchmark. To isolate modality contribution from capacity, we freeze three pretrained encoders (CLIP, wav2vec2, RoBERTa) and train a single small transformer fusion head whose architecture is held constant across all seven possible modality subsets. Averaged over three random seeds, full fusion attains test AUROC 0.876 ± 0.012 , a robust gain over the strongest unimodal baseline (text, 0.827 ± 0.016). On threshold-dependent metrics (accuracy, macro-F1) the gain collapses into seed variance: a single-seed report would have published a spurious $+5.5$ accuracy-point lead that disappeared on re-running. We then audit per-target-community accuracy and observe that no single configuration dominates across target groups, and that the smallest group ($n = 15$) amplifies fusion seed variance to roughly three times that of the text baseline.

Keywords: multimodal learning, hate speech detection, video classification, frozen encoders, transformer fusion, seed variance, fairness audit.

1. Introduction

Hate-speech moderation on video platforms is challenging because hateful intent emerges from the joint configuration of visual context, acoustic delivery, and spoken content. A text classifier misses a slur shouted on the audio; a visual classifier misses harmful framing in the commentary. Hateful Memes [1] pairs image and text; HateMM [2] extends to three modalities on 1,083 BitChute videos with target-community annotations.

We address two gaps in current multimodal hate video work. Published systems typically report single-seed results, which (as we show) overstate threshold-level gains.

Per-community fairness audits on the same trained models are rarely reported alongside aggregate accuracy. Our goal is to quantify per-modality contribution under a controlled ablation and surface whether the gain is uniform across target communities.

Our contribution is methodological. We use a frozen-encoder pipeline (Sec. 3) so the same trainable fusion head runs on any modality subset, isolating *modality information* from model capacity. We train each of the seven non-empty modality subsets under three random seeds and evaluate globally and per target community.

2. Related Work

Multimodal hate detection. Hateful Memes [1] catalysed image-text fusion for moderation but does not capture acoustic delivery. HateMM [2] closes part of this gap and reports late-fusion baselines built on per-modality CLIP and XGBoost classifiers. MM-HSD [3] subsequently introduced richer cross-modal interactions on the same data and remains the closest prior work; however, its evaluation focuses on aggregate accuracy and does not isolate per-modality contribution through an architecture-controlled ablation, nor does it report per-community equity. In the text-only setting, [4] show that specialising general-purpose LLM embeddings transfers well across hate datasets, motivating our use of strong frozen text features rather than training a text encoder from scratch.

Multimodal transformer fusion. Cross-modal attention has matched heavier late-fusion ensembles at lower cost [5]; we adopt the same family at a markedly smaller scale, motivated by HateMM’s modest training size.

Two specific limitations in this prior work inform our design. Single-seed reporting can hide substantial training-time variance on small benchmarks, and aggregate accuracy obscures per-community equity. We address both with a 7-row architecture-controlled ablation under three seeds and a per-community audit.

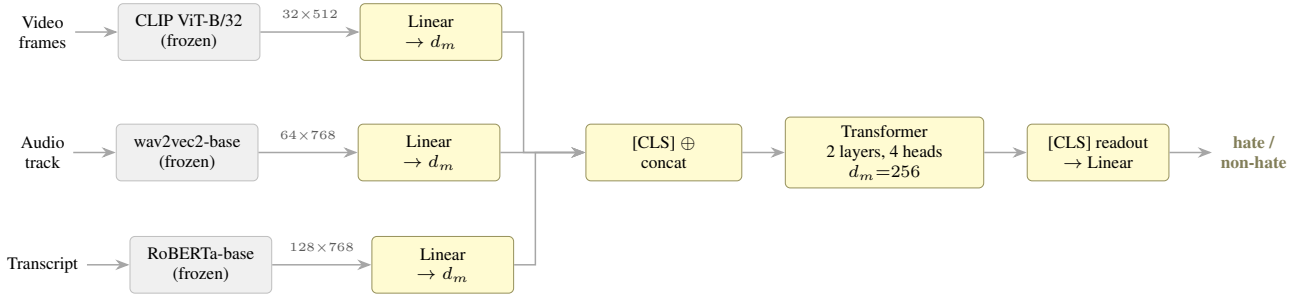


Figure 1. Pipeline architecture. Frozen encoders (grey) produce per-modality feature sequences; trainable layers (gold) project them into a common $d_m=256$ space, prepend a [CLS] token, fuse via a small transformer encoder, and emit a binary prediction. The same head supports any non-empty modality subset by omitting the corresponding input branch.

3. Method

Preprocessing. For each video we extract three streams. We sample 32 RGB frames at 1 fps with center cropping and re-size to 224×224 . The audio track is decoded with `ffmpeg` to 16 kHz mono PCM; videos lacking audio are replaced by a 0.5 s silent placeholder so downstream tensors are uniform. We obtain English transcripts with faster-whisper running the `small` model [6].

Frozen feature extraction. Three pretrained encoders, all kept frozen, produce per-modality feature sequences. Frames pass through the CLIP ViT-B/32 vision tower [7], producing 32 tokens in $\mathbb{R}^{512}$. Audio is encoded with wav2vec2-base [8]; the resulting hidden states are mean-pooled into 64 fixed-length tokens in $\mathbb{R}^{768}$ to bound memory for variable-length inputs. Transcripts pass through RoBERTa-base [9] truncated at 128 tokens, producing token-level embeddings in $\mathbb{R}^{768}$. We cache all features to disk so every later experiment is a cheap tensor read.

Why frozen encoders. With only ~ 865 training videos, finetuning $\sim 300\text{M}$ encoder parameters would risk overfitting. More importantly, freezing means the only trainable component across our seven ablation rows is the fusion head, so the comparison isolates modality information from architectural capacity.

Fusion transformer. As shown in Fig. 1, per-modality features are linearly projected to a common 256-dimensional space and tagged with a learned modality identifier. A trainable [CLS] token is prepended, and the concatenated sequence is processed by a pre-norm transformer encoder [10] with two layers and four attention heads, GELU activations, and dropout 0.1. The [CLS] output is fed to a linear binary classifier. The same module supports any non-empty modality subset by omitting the corresponding embedding sequence, which is how we obtain unimodal and pairwise baselines that share the exact same architecture as the full fusion model. The trainable component totals

$\sim 1.5\text{M}$ parameters.

4. Validation

Dataset and splits. HateMM [2] contains 1,083 BitChute videos (431 hate, 652 non-hate) released under CC BY 4.0, each with a binary label and target-community annotation. Preprocessing yielded 1,081 usable videos. We split stratified by label into 80/10/10 train/val/test (863/109/109) with a fixed split seed. HateMM does not ship canonical splits (the original authors report 5-fold cross-validation), so we use a fixed split shared across all 21 runs of our ablation to keep the held-out test set identical.

Training. All models are optimised with AdamW [11] at learning rate 10^{-4} , weight decay 10^{-2} , batch size 16, using binary cross-entropy. We train for up to 20 epochs with early stopping on validation macro-F1 (patience 5). Each modality configuration is trained under three random seeds {42, 1337, 2025}, with the data split held fixed across seeds. We report mean $\pm$ standard deviation across seeds throughout. Runs use a single NVIDIA V100 GPU; full training across all 21 runs completes in approximately four hours.

Modality ablation. Tab. 1 reports test-set metrics for all seven non-empty modality subsets. AUROC rises monotonically with the number of fused modalities, from 0.77 for audio alone to 0.88 for full fusion, and the gap between full fusion and each unimodal baseline exceeds twice the combined standard deviation. Among pairs, video-and-text leads on AUROC (0.86 ± 0.01), narrowly ahead of video-and-audio (0.85 ± 0.01) and audio-and-text (0.84 ± 0.00). On threshold-dependent metrics the picture inverts: full fusion ties text-only at 0.79 accuracy and 0.78 macro-F1, with markedly higher per-seed variance (accuracy std 0.05 versus 0.02 for text). Fusion improves probability ranking robustly but does not improve fixed-threshold prediction once seed variance is reported. Our first single-seed run

MODS	ACC	F1	PREC	REC	AUROC
V	.72±.03	.71±.02	.64±.06	.68±.06	.81±.01
A	.67±.02	.67±.03	.56±.02	.71±.05	.77±.03
T	.79±.02	.78±.02	.75±.06	.73±.08	.83±.02
V+A	.77±.01	.76±.00	.71±.04	.71±.08	.85±.01
V+T	.78±.03	.77±.02	.73±.07	.72±.04	.86±.01
A+T	.76±.02	.75±.02	.71±.05	.69±.08	.84±.00
V+A+T	.79±.05	.78±.05	.74±.08	.74±.08	.88±.01

Table 1. TEST-SET METRICS ON HATEMM ($n = 109$), MEAN $\pm$ STD OVER THREE SEEDS {42, 1337, 2025}. FULL FUSION WINS ROBUSTLY ON AUROC; ON ACCURACY AND MACRO-F1 IT MATCHES THE TEXT-ONLY BASELINE WITH MARKEDLY HIGHER SEED VARIANCE.

COMMUNITY	n	V	A	T	V+A+T
BLACKS	46	.70±.04	.59±.06	.73±.03	.75±.05
OTHERS	39	.74±.09	.68±.11	.85±.05	.84±.01
JEWS	15	.67±.07	.71±.04	.76±.04	.73±.13

Table 2. PER-TARGET-COMMUNITY TEST ACCURACY (GROUPS WITH $n \geq 15$), MEAN $\pm$ STD OVER THREE SEEDS. DIFFERENCES BETWEEN FUSION AND THE STRONGEST UNIMODAL BASELINE LIE WITHIN COMBINED SEED VARIANCE FOR ALL THREE GROUPS.

reported a +5.5 accuracy-point lead for fusion that vanished after averaging over three seeds: a single-seed report would have published a spurious headline.

Per-community audit. Tab. 2 reports per-community accuracy for the three groups with $n \geq 15$ examples. The aggregate picture does not transfer uniformly. Fusion wins narrowly on *Blacks* ($n=46$, +0.022), text wins on *Others* ($n=39$, +0.017), and text wins on *Jews* ($n=15$, +0.022). On *Jews* the fusion seed std (0.13) is over three times the text std (0.04): high-capacity fusion is unstable on the smallest subgroup. On *Others*, accuracy of ~ 0.85 co-exists with macro-F1 of ~ 0.5 for every model, a label-imbalance artefact in which models default to the majority non-hate class. Remaining target categories provide too few test examples for per-group conclusions.

Discussion. Fusion buys a robust ranking gain but no threshold-level gain once seed variance is reported, consistent with a higher-capacity model that sharpens its score distribution without moving the optimal threshold. Fusion is useful for triage that consumes a ranked queue, less useful for autonomous yes/no moderation. Because no recipe dominates per-community, deployment should be matched to use case: full fusion for ranking-quality, video-and-text as a cost-aware option ($\sim 60\%$ of the AUROC lift over text-only without an audio pipeline), and text-only for fixed-threshold binary calls.

Limitations. The test set ($n = 109$) is small. Community-label normalisation is heuristic, conflating intersectional rows with singletons. Wav2vec2 hidden states are mean-pooled to 64 fixed tokens, over-averaging long videos. Whisper degrades on shouting, music, and accents, biasing the text modality. HateMM is sourced from BitChute (hate base rate $\sim 40\%$), so results require recalibration to transfer to mainstream platforms.

5. Conclusion

By holding the trainable architecture fixed across all seven modality subsets and reporting multi-seed variance, we showed that fusion robustly improves AUROC (0.876 ± 0.012 versus 0.827 ± 0.016 for text) while accuracy and macro-F1 gains collapse into seed variance, a result single-seed reporting would have presented as a spurious threshold-level win. The per-community audit found no recipe that dominates across target groups. Small multimodal benchmarks demand controlled ablations and multi-seed reporting before fusion claims can be trusted.

References

- [1] D. Kiela, H. Firooz, A. Mohan, V. Goswami, A. Singh, P. Ringshia, and D. Tugugine, “The hateful memes challenge: Detecting hate speech in multimodal memes,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2020.
- [2] M. Das, R. Raj, P. Saha, B. Mathew, M. Gupta, and A. Mukherjee, “HateMM: A multi-modal dataset for hate video classification,” in *Proc. AAAI Int. Conf. Web and Social Media (ICWSM)*, 2023.
- [3] B. Céspedes-Sarrias, C. Collado-Capell, P. Rodenas-Ruiz, O. Hrynenko, and A. Cavallaro, “MM-HSD: Multi-modal hate speech detection in videos,” in *Proc. ACM Int. Conf. Multimedia (MM)*, 2025.
- [4] V. Cheremietev, Q. L. H. Ngo, C. Y. Kot, A. E. Baia, and A. Cavallaro, “Specializing general-purpose LLM embeddings for implicit hate speech detection,” in *Proc. Int. Workshop on Diffusion of Harmful Content on Online Web (DHOW)*, 2025.
- [5] A. Nagrani, S. Yang, A. Arnab, A. Jansen, C. Schmid, and C. Sun, “Attention bottlenecks for multimodal fusion,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2021.
- [6] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, “Robust speech recognition via large-scale weak supervision,” *arXiv:2212.04356*, 2022.
- [7] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, “Learning transferable visual models from natural language supervision,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2021.
- [8] A. Baevski, Y. Zhou, A. Mohamed, and M. Auli, “wav2vec 2.0: A framework for self-supervised learning of speech representations,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2020.
- [9] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, “RoBERTa: A robustly optimized BERT pretraining approach,” *arXiv:1907.11692*, 2019.
- [10] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2017.
- [11] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2019.